Deep Learning Assignment — Image Classification

Parts I, II, and III: CNN, Regularization, and Transfer Learning

Dataset: Imagenette (10-class subset of ImageNet) and CIFAR-10
Framework: PyTorch Lightning

```
# Installing required packages
!pip install pytorch-lightning torchmetrics matplotlib

# Downloading Imagenette dataset
!wget -q https://s3.amazonaws.com/fast-ai-imageclas/imagenette2-
160.tgz
!tar -xzf imagenette2-160.tgz

import shutil
import os

os.makedirs('/content/drive/MyDrive/checkpoints', exist_ok=True)

shutil.copy('./checkpoints/best_regularized_cnn.ckpt',

'/content/drive/MyDrive/checkpoints/best_regularized_cnn.ckpt')

shutil.copy('./checkpoints/best_scratch_cifar.ckpt',

'/content/drive/MyDrive/checkpoints/best_scratch_cifar.ckpt')

print('Checkpoints saved to Google Drive!')
```

```
Checkpoints saved to Google Drive!
```

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

```
import os
import copy
import numpy as np
import matplotlib.pyplot as plt
```

```python
import torch
import torch.nn as nn
import torch.nn.functional as F
from torch.utils.data import DataLoader, random_split

import torchvision
import torchvision.transforms as transforms
from torchvision.datasets import ImageFolder, CIFAR10

import pytorch_lightning as pl
from pytorch_lightning.callbacks import EarlyStopping, ModelCheckpoint
from torchmetrics.classification import MulticlassAccuracy

# Reproducibility
pl.seed_everything(42)
DEVICE = 'cuda' if torch.cuda.is_available() else 'cpu'
print(f'Using device: {DEVICE}')
```

```
INFO:lightning_fabric.utilities.seed:Seed set to 42

Using device: cuda
```

Part I — Basic CNN on Imagenette

Architecture Overview

- **3 convolutional blocks**, each with: Conv2d → BatchNorm → ReLU → MaxPool
- **2 fully connected layers** at the end for classification into 10 classes
- **Early Stopping** via PyTorch Lightning's callback (patience=5)

Input images are resized to **128×128** for speed. The 10 Imagenette classes are: tench, English springer, cassette player, chain saw, church, French horn, garbage truck, gas pump, golf ball, parachute.

```python
# ── Data transformation (no augmentation for Part I) ──────────────────────────
IMG_SIZE = 128
BATCH_SIZE = 64
IMAGENETTE_PATH = './imagenette2-160'

base_transform = transforms.Compose([
    transforms.Resize((IMG_SIZE, IMG_SIZE)),
    transforms.ToTensor(),
    transforms.Normalize(mean=[0.485, 0.456, 0.406],
                         std=[0.229, 0.224, 0.225]),
])

train_dataset = ImageFolder(root=os.path.join(IMAGENETTE_PATH,
```

```
'train'),
                              transform=base_transform)
val_dataset   = ImageFolder(root=os.path.join(IMAGENETTE_PATH, 'val'),
                              transform=base_transform)

# Using val set as both validation and test
val_size  = int(0.5 * len(val_dataset))
test_size = len(val_dataset) - val_size
val_data, test_data = random_split(val_dataset, [val_size, test_size])

train_loader = DataLoader(train_dataset, batch_size=BATCH_SIZE,
shuffle=True,  num_workers=2, pin_memory=True)
val_loader   = DataLoader(val_data,      batch_size=BATCH_SIZE,
shuffle=False, num_workers=2, pin_memory=True)
test_loader  = DataLoader(test_data,     batch_size=BATCH_SIZE,
shuffle=False, num_workers=2, pin_memory=True)

print(f'Train: {len(train_dataset)} | Val: {len(val_data)} | Test:
{len(test_data)}')
```

```
Train: 9469 | Val: 1962 | Test: 1963
```

```
# ── CNN Architecture ──
class BasicCNN(pl.LightningModule):
    """
    3-block CNN for 10-class image classification.
    Each block: Conv2d(k=3,p=1) -> BatchNorm -> ReLU -> MaxPool(2)
    Final head: Flatten -> FC(512) -> ReLU -> Dropout -> FC(10)
    """
    def __init__(self, num_classes=10, lr=1e-3, dropout_p=0.0):
        super().__init__()
        self.save_hyperparameters()
        self.lr = lr
        self.train_losses, self.val_losses = [], []

        self.features = nn.Sequential(
            # Block 1: 3 -> 32 channels, 128 -> 64
            nn.Conv2d(3, 32, kernel_size=3, padding=1),
            nn.BatchNorm2d(32),
            nn.ReLU(inplace=True),
            nn.MaxPool2d(2, 2),

            # Block 2: 32 -> 64 channels, 64 -> 32
            nn.Conv2d(32, 64, kernel_size=3, padding=1),
            nn.BatchNorm2d(64),
            nn.ReLU(inplace=True),
            nn.MaxPool2d(2, 2),

            # Block 3: 64 -> 128 channels, 32 -> 16
            nn.Conv2d(64, 128, kernel_size=3, padding=1),
```

```
            nn.BatchNorm2d(128),
            nn.ReLU(inplace=True),
            nn.MaxPool2d(2, 2),
        )
        # After 3 x MaxPool(2): 128 -> 16; feature map = 128 * 16 * 16
        self.classifier = nn.Sequential(
            nn.Flatten(),
            nn.Linear(128 * 16 * 16, 512),
            nn.ReLU(inplace=True),
            nn.Dropout(p=dropout_p),
            nn.Linear(512, num_classes),
        )
        self.accuracy = MulticlassAccuracy(num_classes=num_classes)

    def forward(self, x):
        return self.classifier(self.features(x))

    def _shared_step(self, batch):
        x, y = batch
        logits = self(x)
        loss = F.cross_entropy(logits, y)
        acc  = self.accuracy(logits, y)
        return loss, acc

    def training_step(self, batch, batch_idx):
        loss, acc = self._shared_step(batch)
        self.log('train_loss', loss, on_epoch=True, prog_bar=True)
        self.log('train_acc',  acc,  on_epoch=True, prog_bar=True)
        return loss

    def validation_step(self, batch, batch_idx):
        loss, acc = self._shared_step(batch)
        self.log('val_loss', loss, on_epoch=True, prog_bar=True)
        self.log('val_acc',  acc,  on_epoch=True, prog_bar=True)

    def test_step(self, batch, batch_idx):
        loss, acc = self._shared_step(batch)
        self.log('test_loss', loss)
        self.log('test_acc',  acc)

    def configure_optimizers(self):
        optimizer = torch.optim.Adam(self.parameters(), lr=self.lr)
        scheduler = torch.optim.lr_scheduler.ReduceLROnPlateau(
            optimizer, mode='min', factor=0.5, patience=3
        )
        return {'optimizer': optimizer,
                'lr_scheduler': {'scheduler': scheduler, 'monitor':
'val_loss'}}
```

```python
print('Model defined.')
model_v1 = BasicCNN(num_classes=10, lr=1e-3, dropout_p=0.0)
print(model_v1)
```

```
Model defined.
BasicCNN(
  (features): Sequential(
    (0): Conv2d(3, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1))
    (1): BatchNorm2d(32, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (2): ReLU(inplace=True)
    (3): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1,
ceil_mode=False)
    (4): Conv2d(32, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1))
    (5): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (6): ReLU(inplace=True)
    (7): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1,
ceil_mode=False)
    (8): Conv2d(64, 128, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1))
    (9): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (10): ReLU(inplace=True)
    (11): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1,
ceil_mode=False)
  )
  (classifier): Sequential(
    (0): Flatten(start_dim=1, end_dim=-1)
    (1): Linear(in_features=32768, out_features=512, bias=True)
    (2): ReLU(inplace=True)
    (3): Dropout(p=0.0, inplace=False)
    (4): Linear(in_features=512, out_features=10, bias=True)
  )
  (accuracy): MulticlassAccuracy()
)
```

```python
# ── Training — Part I ───
early_stop_cb = EarlyStopping(monitor='val_loss', patience=5,
mode='min', verbose=True)
ckpt_cb_v1    = ModelCheckpoint(monitor='val_loss', save_top_k=1,
                                filename='best_basic_cnn',
dirpath='./checkpoints')

trainer_v1 = pl.Trainer(
    max_epochs=30,
    callbacks=[early_stop_cb, ckpt_cb_v1],
    accelerator='auto',
```

```
    log_every_n_steps=10,
)

trainer_v1.fit(model_v1, train_loader, val_loader)
test_result_v1 = trainer_v1.test(model_v1, test_loader,
ckpt_path='best')
print('\nPart I Test Results:', test_result_v1)
```

```
INFO:pytorch_lightning.utilities.rank_zero:GPU available: True (cuda),
used: True
INFO:pytorch_lightning.utilities.rank_zero:TPU available: False,
using: 0 TPU cores
INFO:pytorch_lightning.utilities.rank_zero:💡 Tip: For seamless cloud
logging and experiment tracking, try installing
[litlogger](https://pypi.org/project/litlogger/) to enable LitLogger,
which logs metrics and artifacts automatically to the Lightning
Experiments platform.
INFO:pytorch_lightning.accelerators.cuda:LOCAL_RANK: 0 -
CUDA_VISIBLE_DEVICES: [0]
```

	Name	Type	Params	Mode	FLOPs
0	features	Sequential	93.7 K	train	0
1	classifier	Sequential	16.8 M	train	0
2	accuracy	MulticlassAccuracy	0	train	0

```
Trainable params: 16.9 M

Non-trainable params: 0

Total params: 16.9 M

Total estimated model params size (MB): 67

Modules in train mode: 20

Modules in eval mode: 0

Total FLOPs: 0
```

{"model_id":"cfe2b3fe50534dff9427debe7c8865a6","version_major":2,"version_minor":0}

```
/usr/local/lib/python3.12/dist-packages/pytorch_lightning/utilities/
_pytree.py:21: `isinstance(treespec, LeafSpec)`
is deprecated, use `isinstance(treespec, TreeSpec) and
treespec.is_leaf()` instead.
```

```
INFO:pytorch_lightning.callbacks.early_stopping:Metric val_loss
improved. New best score: 1.516
INFO:pytorch_lightning.callbacks.early_stopping:Metric val_loss
improved by 0.148 >= min_delta = 0.0. New best score: 1.368
INFO:pytorch_lightning.callbacks.early_stopping:Metric val_loss
improved by 0.041 >= min_delta = 0.0. New best score: 1.327
INFO:pytorch_lightning.callbacks.early_stopping:Metric val_loss
improved by 0.198 >= min_delta = 0.0. New best score: 1.129
INFO:pytorch_lightning.callbacks.early_stopping:Metric val_loss
improved by 0.013 >= min_delta = 0.0. New best score: 1.116
INFO:pytorch_lightning.callbacks.early_stopping:Monitored metric
val_loss did not improve in the last 5 records. Best score: 1.116.
Signaling Trainer to stop.
```

```
INFO:pytorch_lightning.utilities.rank_zero:Restoring states from the
checkpoint path at /content/checkpoints/best_basic_cnn.ckpt
INFO:pytorch_lightning.accelerators.cuda:LOCAL_RANK: 0 -
CUDA_VISIBLE_DEVICES: [0]
INFO:pytorch_lightning.utilities.rank_zero:Loaded model weights from
the checkpoint at /content/checkpoints/best_basic_cnn.ckpt
```

{"model_id":"4d3b18e3157942d2842671bc0c52007e","version_major":2,"version_minor":0}

Test metric	DataLoader 0
test_acc	0.6519144773483276
test_loss	1.119805097579956

```
Part I Test Results: [{'test_loss': 1.119805097579956, 'test_acc':
0.6519144773483276}]
```

```
# ── Plot Part I — Training & Validation Loss ──
def get_metric(trainer, key):
    """Extract per-epoch metric from Lightning trainer logs."""
    metrics = trainer.logged_metrics

    history = []
    for cb in trainer.loggers:
        if hasattr(cb, 'experiment') and hasattr(cb.experiment,
'metrics'):
            history = [m[key] for m in cb.experiment.metrics if key in
m]
    return history
```

```python
# Re-training with CSVLogger to capture epoch histories easily
from pytorch_lightning.loggers import CSVLogger
import pandas as pd

logger_v1 = CSVLogger('.', name='logs_v1')
trainer_v1b = pl.Trainer(
    max_epochs=30,
    callbacks=[
        EarlyStopping(monitor='val_loss', patience=5, mode='min'),
        ModelCheckpoint(monitor='val_loss', save_top_k=1,
                        filename='best_basic_cnn',
dirpath='./checkpoints'),
    ],
    accelerator='auto',
    logger=logger_v1,
    log_every_n_steps=10,
)
model_v1b = BasicCNN(num_classes=10, lr=1e-3, dropout_p=0.0)
trainer_v1b.fit(model_v1b, train_loader, val_loader)
result_v1 = trainer_v1b.test(model_v1b, test_loader, ckpt_path='best')

# Read and plot logs
log_path = logger_v1.experiment.metrics_file_path
df_v1 = pd.read_csv(log_path)

train_loss_v1 = df_v1.dropna(subset=['train_loss_epoch'])
['train_loss_epoch'].values
val_loss_v1   = df_v1.dropna(subset=['val_loss'])['val_loss'].values
epochs_v1 = range(1, len(train_loss_v1) + 1)

fig, axes = plt.subplots(1, 2, figsize=(12, 4))
axes[0].plot(epochs_v1, train_loss_v1, 'b-o', label='Train Loss')
axes[0].plot(range(1, len(val_loss_v1)+1), val_loss_v1, 'r-o',
label='Val Loss')
axes[0].set_title('Part I: Training vs Validation Loss')
axes[0].set_xlabel('Epoch')
axes[0].set_ylabel('Loss')
axes[0].legend()
axes[0].grid(True)

train_acc_v1 = df_v1.dropna(subset=['train_acc_epoch'])
['train_acc_epoch'].values
val_acc_v1   = df_v1.dropna(subset=['val_acc'])['val_acc'].values
axes[1].plot(range(1, len(train_acc_v1)+1), train_acc_v1, 'b-o',
label='Train Acc')
axes[1].plot(range(1, len(val_acc_v1)+1), val_acc_v1, 'r-o',
label='Val Acc')
axes[1].set_title('Part I: Training vs Validation Accuracy')
axes[1].set_xlabel('Epoch')
```

```
axes[1].set_ylabel('Accuracy')
axes[1].legend()
axes[1].grid(True)

plt.tight_layout()
plt.savefig('part1_training_curves.png', dpi=150, bbox_inches='tight')
plt.show()
print(f"\nPart I Final Test Accuracy: {result_v1[0]['test_acc']:.4f}")
```

```
INFO:pytorch_lightning.utilities.rank_zero:GPU available: True (cuda),
used: True
INFO:pytorch_lightning.utilities.rank_zero:TPU available: False,
using: 0 TPU cores
INFO:pytorch_lightning.utilities.rank_zero:□ Tip: For seamless cloud
logging and experiment tracking, try installing
[litlogger](https://pypi.org/project/litlogger/) to enable LitLogger,
which logs metrics and artifacts automatically to the Lightning
Experiments platform.
/usr/local/lib/python3.12/dist-packages/pytorch_lightning/callbacks/
model_checkpoint.py:881: Checkpoint directory /content/checkpoints
exists and is not empty.
INFO:pytorch_lightning.accelerators.cuda:LOCAL_RANK: 0 -
CUDA_VISIBLE_DEVICES: [0]
```

	Name	Type	Params	Mode	FLOPs
0	features	Sequential	93.7 K	train	0
1	classifier	Sequential	16.8 M	train	0
2	accuracy	MulticlassAccuracy	0	train	0

```
Trainable params: 16.9 M

Non-trainable params: 0

Total params: 16.9 M

Total estimated model params size (MB): 67

Modules in train mode: 20

Modules in eval mode: 0

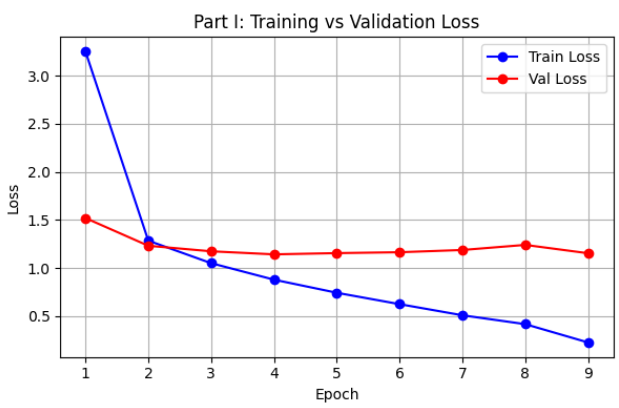
Total FLOPs: 0
```

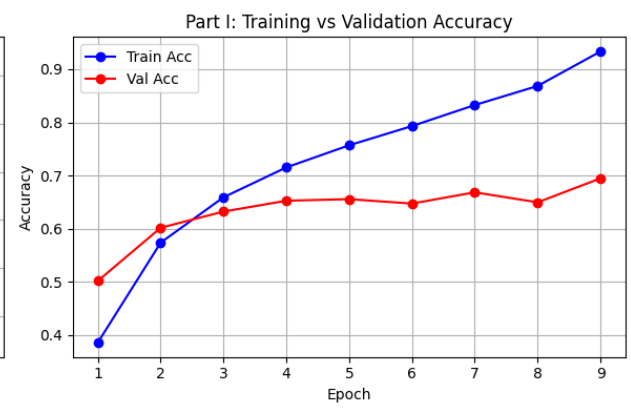
{"model_id":"bc4cd5db660848d79c8fb955343a9419","version_major":2,"version_minor":0}

```
INFO:pytorch_lightning.utilities.rank_zero:Restoring states from the
checkpoint path at /content/checkpoints/best_basic_cnn-v1.ckpt
INFO:pytorch_lightning.accelerators.cuda:LOCAL_RANK: 0 -
CUDA_VISIBLE_DEVICES: [0]
INFO:pytorch_lightning.utilities.rank_zero:Loaded model weights from
the checkpoint at /content/checkpoints/best_basic_cnn-v1.ckpt
```

{"model_id":"5b68ca25d58845adaa16280594eb96b0","version_major":2,"version_minor":0}

Test metric	DataLoader 0
test_acc	0.6530743837356567
test_loss	1.0972023010253906






```
Part I Final Test Accuracy: 0.6531
```

Part I Results

Architecture Summary: | Layer | Output Size | Details | |---|---|---| | Input | 3×128×128 | RGB image | | Conv Block 1 | 32×64×64 | Conv(3,32), BN, ReLU, MaxPool | | Conv Block 2 | 64×32×32 | Conv(32,64), BN, ReLU, MaxPool | | Conv Block 3 | 128×16×16 | Conv(64,128), BN, ReLU, MaxPool | | FC 1 | 512 | Linear, ReLU | | FC 2 (Output) | 10 | Linear (logits) |

- **Optimizer:** Adam (lr=1e-3), ReduceLROnPlateau scheduler
- **Early Stopping:** patience=5 on `val_loss`
- **Batch size:** 64

Part II — Regularization

Two forms of regularization:

1. **Data Augmentation** — Random horizontal flip, random crop, color jitter
2. **Dropout** — p=0.5 in the classifier head

These help the model generalize by making it harder to memorize training samples.

```
# — Augmented Transform —
aug_transform = transforms.Compose([
    transforms.Resize((IMG_SIZE + 16, IMG_SIZE + 16)),
    transforms.RandomCrop(IMG_SIZE),
    transforms.RandomHorizontalFlip(p=0.5),
    transforms.ColorJitter(brightness=0.3, contrast=0.3,
saturation=0.3, hue=0.1),
    transforms.ToTensor(),
    transforms.Normalize(mean=[0.485, 0.456, 0.406],
                         std=[0.229, 0.224, 0.225]),
])

# Training with augmentation; val/test keep base transform
train_aug_dataset = ImageFolder(root=os.path.join(IMAGENETTE_PATH,
'train'),
                                transform=aug_transform)
train_aug_loader  = DataLoader(train_aug_dataset,
batch_size=BATCH_SIZE,
                               shuffle=True, num_workers=2,
pin_memory=True)

print('Augmented train loader ready.')
```

```
Augmented train loader ready.
```

```
# — Train regularized model —
logger_v2 = CSVLogger('.', name='logs_v2')
trainer_v2 = pl.Trainer(
    max_epochs=50,
    callbacks=[
        EarlyStopping(monitor='val_loss', patience=7, mode='min'),
#changed patience to 7
        ModelCheckpoint(monitor='val_loss', save_top_k=1,
                        filename='best_regularized_cnn',
dirpath='./checkpoints'),
    ],
    accelerator='auto',
    logger=logger_v2,
    log_every_n_steps=10,
)
```

```
model_v2 = BasicCNN(num_classes=10, lr=1e-3, dropout_p=0.3)  # dropout=0.3
trainer_v2.fit(model_v2, train_aug_loader, val_loader)        # augmented train data
result_v2 = trainer_v2.test(model_v2, test_loader, ckpt_path='best')
print('\nPart II Test Results:', result_v2)
```

```
INFO:pytorch_lightning.utilities.rank_zero:GPU available: True (cuda), used: True
INFO:pytorch_lightning.utilities.rank_zero:TPU available: False, using: 0 TPU cores
INFO:pytorch_lightning.utilities.rank_zero:💡 Tip: For seamless cloud logging and experiment tracking, try installing [litlogger](https://pypi.org/project/litlogger/) to enable LitLogger, which logs metrics and artifacts automatically to the Lightning Experiments platform.
/usr/local/lib/python3.12/dist-packages/pytorch_lightning/callbacks/model_checkpoint.py:881: Checkpoint directory /content/checkpoints exists and is not empty.
INFO:pytorch_lightning.accelerators.cuda:LOCAL_RANK: 0 - CUDA_VISIBLE_DEVICES: [0]
```

	Name	Type	Params	Mode	FLOPs
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1	classifier	Sequential	16.8 M	train	0
2	accuracy	MulticlassAccuracy	0	train	0

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Modules in train mode: 20

Modules in eval mode: 0

Total FLOPs: 0
```

{"model_id":"4078f38f3fea45eeba8911cca923991d","version_major":2,"version_minor":0}

```
/usr/local/lib/python3.12/dist-packages/pytorch_lightning/utilities/_pytree.py:21: `isinstance(treespec, LeafSpec)`
is deprecated, use `isinstance(treespec, TreeSpec) and
treespec.is_leaf()` instead.
```

```
INFO:pytorch_lightning.utilities.rank_zero:Restoring states from the
checkpoint path at /content/checkpoints/best_regularized_cnn.ckpt
INFO:pytorch_lightning.accelerators.cuda:LOCAL_RANK: 0 -
CUDA_VISIBLE_DEVICES: [0]
INFO:pytorch_lightning.utilities.rank_zero:Loaded model weights from
the checkpoint at /content/checkpoints/best_regularized_cnn.ckpt
```

{"model_id":"23ca15b125cf47d0be8caded1d9baf5c","version_major":2,"version_minor":0}

Test metric	DataLoader 0
test_acc	0.781903862953186
test_loss	0.6728731393814087

```
Part II Test Results: [{'test_loss': 0.6728731393814087, 'test_acc':
0.781903862953186}]
```

```
# ── Comparison Plot: With vs Without Regularization ──
df_v2 = pd.read_csv(logger_v2.experiment.metrics_file_path)

train_loss_v2 = df_v2.dropna(subset=['train_loss_epoch'])
['train_loss_epoch'].values
val_loss_v2   = df_v2.dropna(subset=['val_loss'])['val_loss'].values
val_acc_v2    = df_v2.dropna(subset=['val_acc'])['val_acc'].values

fig, axes = plt.subplots(1, 2, figsize=(14, 5))

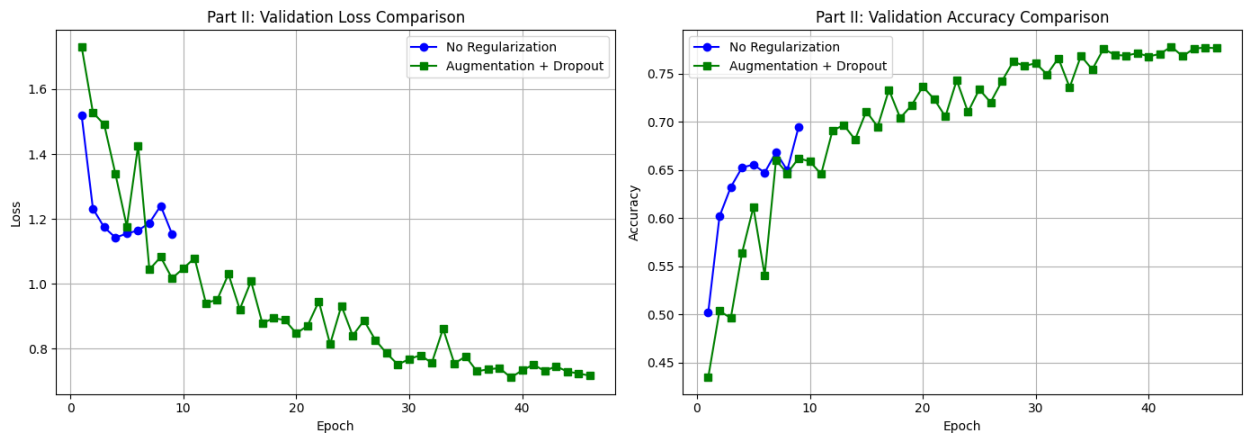
# Val Loss comparison
axes[0].plot(range(1, len(val_loss_v1)+1), val_loss_v1, 'b-o',
label='No Regularization')
axes[0].plot(range(1, len(val_loss_v2)+1), val_loss_v2, 'g-s',
label='Augmentation + Dropout')
axes[0].set_title('Part II: Validation Loss Comparison')
axes[0].set_xlabel('Epoch')
axes[0].set_ylabel('Loss')
axes[0].legend()
axes[0].grid(True)

# Val Accuracy comparison
axes[1].plot(range(1, len(val_acc_v1)+1), val_acc_v1, 'b-o', label='No
Regularization')
axes[1].plot(range(1, len(val_acc_v2)+1), val_acc_v2, 'g-s',
label='Augmentation + Dropout')
```

```
axes[1].set_title('Part II: Validation Accuracy Comparison')
axes[1].set_xlabel('Epoch')
axes[1].set_ylabel('Accuracy')
axes[1].legend()
axes[1].grid(True)

plt.tight_layout()
plt.savefig('part2_regularization_comparison.png', dpi=150,
bbox_inches='tight')
plt.show()

print(f"\nNo Regularization  — Test Accuracy: {result_v1[0]
['test_acc']:.4f}")
print(f"With Regularization — Test Accuracy: {result_v2[0]
['test_acc']:.4f}")
```




```
No Regularization  — Test Accuracy: 0.6531
With Regularization — Test Accuracy: 0.7819
```

Part II: Regularization Description

Data Augmentation applied (training set only): | Technique | Parameters | Purpose | |---|---|---| | RandomCrop | 128×128 from 144×144 | Positional invariance | | RandomHorizontalFlip | p=0.5 | Mirror symmetry | | ColorJitter | brightness/contrast/saturation=0.3, hue=0.1 | Lighting robustness |

Dropout: p=0.5 applied after the first fully-connected layer.

Expected outcome: The regularized model should show a smaller gap between training and validation performance, indicating better generalization.

Part III — Transfer Learning Strategy:

Saved the weights from our best Imagenette model (Part II)

Created the same architecture adapted for CIFAR-10 (also 10 classes)

Trained it from scratch on CIFAR-10 → baseline

Trained it initialized with Imagenette weights (fine-tuning) → transfer learning

Compared convergence speed and final accuracy **bold text** bold text

> Add blockquote

The convolutional layers learn low-level features (edges, textures) that transfer across datasets. Only the high-level representations differ.

> Add blockquote

```
# ── CIFAR-10 DataLoaders ──
# CIFAR-10 images are 32×32 — resized to 128×128 to match our CNN
input
cifar_transform = transforms.Compose([
    transforms.Resize((IMG_SIZE, IMG_SIZE)),
    transforms.ToTensor(),
    transforms.Normalize(mean=[0.4914, 0.4822, 0.4465],   # CIFAR-10
stats
                         std=[0.2023, 0.1994, 0.2010]),
])

cifar_aug_transform = transforms.Compose([
    transforms.Resize((IMG_SIZE + 16, IMG_SIZE + 16)),
    transforms.RandomCrop(IMG_SIZE),
    transforms.RandomHorizontalFlip(),
    transforms.ToTensor(),
    transforms.Normalize(mean=[0.4914, 0.4822, 0.4465],
                         std=[0.2023, 0.1994, 0.2010]),
])

cifar_train = CIFAR10(root='./data', train=True,  download=True,
transform=cifar_aug_transform)
cifar_test  = CIFAR10(root='./data', train=False, download=True,
transform=cifar_transform)

# Split train into train/val
n_val = int(0.1 * len(cifar_train))
cifar_train_data, cifar_val_data = random_split(cifar_train,
[len(cifar_train)-n_val, n_val])

cifar_train_loader = DataLoader(cifar_train_data,
batch_size=BATCH_SIZE, shuffle=True,  num_workers=2)
cifar_val_loader   = DataLoader(cifar_val_data,
batch_size=BATCH_SIZE, shuffle=False, num_workers=2)
cifar_test_loader  = DataLoader(cifar_test,
batch_size=BATCH_SIZE, shuffle=False, num_workers=2)

print(f'CIFAR-10 — Train: {len(cifar_train_data)} | Val:
{len(cifar_val_data)} | Test: {len(cifar_test)}')
```

```
100%|██████████| 170M/170M [00:13<00:00, 12.8MB/s]

CIFAR-10 — Train: 45000 | Val: 5000 | Test: 10000
```

```python
# ── 3A: Train from scratch on CIFAR-10 ──
logger_v3a = CSVLogger('.', name='logs_v3_scratch')

model_scratch = BasicCNN(num_classes=10, lr=1e-3, dropout_p=0.3)

trainer_v3a = pl.Trainer(
    max_epochs=60, #patience and epoch changed
    callbacks=[
        EarlyStopping(monitor='val_loss', patience=8, mode='min'),
        ModelCheckpoint(monitor='val_loss', save_top_k=1,
                        filename='best_scratch_cifar',
dirpath='./checkpoints'),
    ],
    accelerator='auto',
    logger=logger_v3a,
    log_every_n_steps=10,
)

trainer_v3a.fit(model_scratch, cifar_train_loader, cifar_val_loader)
result_scratch = trainer_v3a.test(model_scratch, cifar_test_loader,
ckpt_path='best')
print('\nScratch CIFAR-10 Test Results:', result_scratch)
```

```
INFO:pytorch_lightning.utilities.rank_zero:GPU available: True (cuda),
used: True
INFO:pytorch_lightning.utilities.rank_zero:TPU available: False,
using: 0 TPU cores
INFO:pytorch_lightning.utilities.rank_zero:💡 Tip: For seamless cloud
logging and experiment tracking, try installing
[litlogger](https://pypi.org/project/litlogger/) to enable LitLogger,
which logs metrics and artifacts automatically to the Lightning
Experiments platform.
/usr/local/lib/python3.12/dist-packages/pytorch_lightning/callbacks/
model_checkpoint.py:881: Checkpoint directory /content/checkpoints
exists and is not empty.
INFO:pytorch_lightning.accelerators.cuda:LOCAL_RANK: 0 -
CUDA_VISIBLE_DEVICES: [0]
```

	Name	Type	Params	Mode	FLOPs
0	features	Sequential	93.7 K	train	0
1	classifier	Sequential	16.8 M	train	0
2	accuracy	MulticlassAccuracy	0	train	0

```
Trainable params: 16.9 M

Non-trainable params: 0

Total params: 16.9 M

Total estimated model params size (MB): 67

Modules in train mode: 20

Modules in eval mode: 0

Total FLOPs: 0
```

{"model_id":"a1359347c1eb48279507e5540c1a8d49","version_major":2,"version_minor":0}

```
/usr/local/lib/python3.12/dist-packages/pytorch_lightning/utilities/
_pytree.py:21: `isinstance(treespec, LeafSpec)`
is deprecated, use `isinstance(treespec, TreeSpec) and
treespec.is_leaf()` instead.

INFO:pytorch_lightning.utilities.rank_zero:`Trainer.fit` stopped:
`max_epochs=60` reached.



INFO:pytorch_lightning.utilities.rank_zero:Restoring states from the
checkpoint path at /content/checkpoints/best_scratch_cifar.ckpt
INFO:pytorch_lightning.accelerators.cuda:LOCAL_RANK: 0 -
CUDA_VISIBLE_DEVICES: [0]
INFO:pytorch_lightning.utilities.rank_zero:Loaded model weights from
the checkpoint at /content/checkpoints/best_scratch_cifar.ckpt
```

{"model_id":"3f1df63b3ea54f88a3ab4bb06dca2858","version_major":2,"version_minor":0}

Test metric	DataLoader 0
test_acc	0.7610042095184326
test_loss	0.6896587610244751

```
Scratch CIFAR-10 Test Results: [{'test_loss': 0.6896587610244751,
'test_acc': 0.7610042095184326}]

# ── 3B: Fine-tune from Imagenette pre-trained weights ──
# Load best Imagenette checkpoint and copy feature extractor weights
```

```
imagenette_ckpt = imagenette_ckpt =
'/content/drive/MyDrive/checkpoints/best_regularized_cnn.ckpt'


# Build model with same architecture (10 classes — both datasets have
10)
model_finetune = BasicCNN(num_classes=10, lr=3e-4, dropout_p=0.3)  #
lowering lr and p for fine-tuning

# Load Imagenette weights (strict=False ignores mismatched keys if
any)
pretrained_state = torch.load(imagenette_ckpt, map_location='cpu')
['state_dict']
model_finetune.load_state_dict(pretrained_state, strict=True)
print('Imagenette pre-trained weights loaded successfully.')

for param in model_finetune.features.parameters():
    param.requires_grad = True   # allow full fine-tuning (unfreeze
all)

logger_v3b = CSVLogger('.', name='logs_v3_finetune')

trainer_v3b = pl.Trainer(
    max_epochs=60,
    callbacks=[
        EarlyStopping(monitor='val_loss', patience=8, mode='min'),
        ModelCheckpoint(monitor='val_loss', save_top_k=1,
                        filename='best_finetune_cifar',
dirpath='./checkpoints'),
    ],
    accelerator='auto',
    logger=logger_v3b,
    log_every_n_steps=10,
)

trainer_v3b.fit(model_finetune, cifar_train_loader, cifar_val_loader)
result_finetune = trainer_v3b.test(model_finetune, cifar_test_loader,
ckpt_path='best')
print('\nFine-tuned CIFAR-10 Test Results:', result_finetune)
```

```
INFO:pytorch_lightning.utilities.rank_zero:GPU available: True (cuda),
used: True
INFO:pytorch_lightning.utilities.rank_zero:TPU available: False,
using: 0 TPU cores
INFO:pytorch_lightning.utilities.rank_zero:💡 Tip: For seamless cloud
logging and experiment tracking, try installing
[litlogger](https://pypi.org/project/litlogger/) to enable LitLogger,
which logs metrics and artifacts automatically to the Lightning
Experiments platform.
/usr/local/lib/python3.12/dist-packages/pytorch_lightning/callbacks/
```

```
model_checkpoint.py:881: Checkpoint directory /content/checkpoints
exists and is not empty.
INFO:pytorch_lightning.accelerators.cuda:LOCAL_RANK: 0 -
CUDA_VISIBLE_DEVICES: [0]

Imagenette pre-trained weights loaded successfully.
```

	Name	Type	Params	Mode	FLOPs
0	features	Sequential	93.7 K	train	0
1	classifier	Sequential	16.8 M	train	0
2	accuracy	MulticlassAccuracy	0	train	0

```
Trainable params: 16.9 M

Non-trainable params: 0

Total params: 16.9 M

Total estimated model params size (MB): 67

Modules in train mode: 20

Modules in eval mode: 0

Total FLOPs: 0
```

{"model_id":"8ff0147a32d648b39103c423e8e2454f","version_major":2,"version_minor":0}

```
/usr/local/lib/python3.12/dist-packages/pytorch_lightning/utilities/
_pytree.py:21: `isinstance(treespec, LeafSpec)`
is deprecated, use `isinstance(treespec, TreeSpec) and
treespec.is_leaf()` instead.



---------------------------------------------------------------------------
FileNotFoundError                         Traceback (most recent call
last)
/tmp/ipykernel_6524/906612669.py in <cell line: 0>()
     29 )
     30
---> 31 trainer_v3b.fit(model_finetune, cifar_train_loader,
cifar_val_loader)
     32 result_finetune = trainer_v3b.test(model_finetune,
cifar_test_loader, ckpt_path='best')
     33 print('\nFine-tuned CIFAR-10 Test Results:', result_finetune)
```

```
/usr/local/lib/python3.12/dist-packages/pytorch_lightning/trainer/
trainer.py in fit(self, model, train_dataloaders, val_dataloaders,
datamodule, ckpt_path, weights_only)
    582         self.training = True
    583         self.should_stop = False
--> 584         call._call_and_handle_interrupt(
    585             self,
    586             self._fit_impl,

/usr/local/lib/python3.12/dist-packages/pytorch_lightning/trainer/
call.py in _call_and_handle_interrupt(trainer, trainer_fn, *args,
**kwargs)
     47         if trainer.strategy.launcher is not None:
     48             return
trainer.strategy.launcher.launch(trainer_fn, *args, trainer=trainer,
**kwargs)
---> 49         return trainer_fn(*args, **kwargs)
     50
     51     except _TunerExitException:

/usr/local/lib/python3.12/dist-packages/pytorch_lightning/trainer/
trainer.py in _fit_impl(self, model, train_dataloaders,
val_dataloaders, datamodule, ckpt_path, weights_only)
    628             model_connected=self.lightning_module is not None,
    629         )
--> 630         self._run(model, ckpt_path=ckpt_path,
weights_only=weights_only)
    631
    632         assert self.state.stopped

/usr/local/lib/python3.12/dist-packages/pytorch_lightning/trainer/
trainer.py in _run(self, model, ckpt_path, weights_only)
   1077         # RUN THE TRAINER
   1078         # ----------------------------
-> 1079         results = self._run_stage()
   1080
   1081         # ----------------------------

/usr/local/lib/python3.12/dist-packages/pytorch_lightning/trainer/
trainer.py in _run_stage(self)
   1121                 self._run_sanity_check()
   1122             with
torch.autograd.set_detect_anomaly(self._detect_anomaly):
-> 1123                 self.fit_loop.run()
   1124             return None
   1125         raise RuntimeError(f"Unexpected state {self.state}")

/usr/local/lib/python3.12/dist-packages/pytorch_lightning/loops/fit_lo
op.py in run(self)
    215             try:
```

```
    216                 self.on_advance_start()
--> 217                 self.advance()
    218                 self.on_advance_end()
    219             except StopIteration:

/usr/local/lib/python3.12/dist-packages/pytorch_lightning/loops/fit_lo
op.py in advance(self)
    463         with
self.trainer.profiler.profile("run_training_epoch"):
    464             assert self._data_fetcher is not None
--> 465             self.epoch_loop.run(self._data_fetcher)
    466
    467     def on_advance_end(self) -> None:

/usr/local/lib/python3.12/dist-packages/pytorch_lightning/loops/traini
ng_epoch_loop.py in run(self, data_fetcher)
    151         while not self.done:
    152             try:
--> 153                 self.advance(data_fetcher)
    154                 self.on_advance_end(data_fetcher)
    155             except StopIteration:

/usr/local/lib/python3.12/dist-packages/pytorch_lightning/loops/traini
ng_epoch_loop.py in advance(self, data_fetcher)
    309         else:
    310             dataloader_iter = None
--> 311             batch, _, __ = next(data_fetcher)
    312             # TODO: we should instead use the batch_idx
returned by the fetcher, however, that will require saving the
    313             # fetcher state so that the batch_idx is correct
after restarting

/usr/local/lib/python3.12/dist-packages/pytorch_lightning/loops/fetche
rs.py in __next__(self)
    132         elif not self.done:
    133             # this will run only when no pre-fetching was
done.
--> 134             batch = super().__next__()
    135         else:
    136             # the iterator is empty

/usr/local/lib/python3.12/dist-packages/pytorch_lightning/loops/fetche
rs.py in __next__(self)
     59         self._start_profiler()
     60         try:
---> 61             batch = next(self.iterator)
     62         except StopIteration:
     63             self.done = True

/usr/local/lib/python3.12/dist-packages/pytorch_lightning/utilities/
```

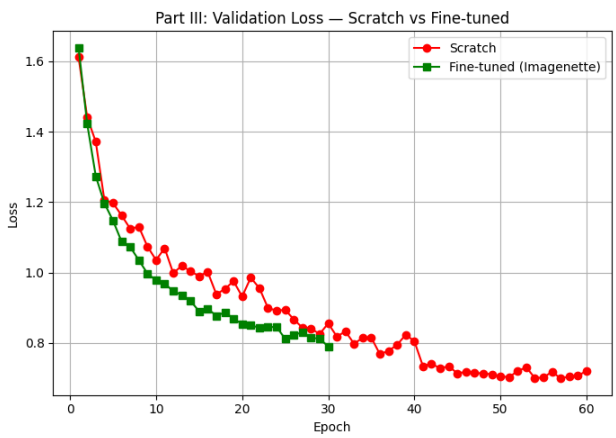
```
combined_loader.py in __next__(self)
    339     def __next__(self) -> _ITERATOR_RETURN:
    340         assert self._iterator is not None
--> 341         out = next(self._iterator)
    342         if isinstance(self._iterator, _Sequential):
    343             return out

/usr/local/lib/python3.12/dist-packages/pytorch_lightning/utilities/
combined_loader.py in __next__(self)
     76         for i in range(n):
     77             try:
---> 78                 out[i] = next(self.iterators[i])
     79             except StopIteration:
     80                 self._consumed[i] = True

/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py
in __next__(self)
    739                 #
TODO(https://github.com/pytorch/pytorch/issues/76750)
    740                 self._reset()  # type: ignore[call-arg]
--> 741             data = self._next_data()
    742             self._num_yielded += 1
    743             if (

/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py
in _next_data(self)
   1522                     "Invalid iterator state: shutdown or no
outstanding tasks when fetching next data"
   1523                 )
-> 1524             idx, data = self._get_data()
   1525             self._tasks_outstanding -= 1
   1526             if self._dataset_kind == _DatasetKind.Iterable:

/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py
in _get_data(self)
   1481         else:
   1482             while True:
-> 1483                 success, data = self._try_get_data()
   1484                 if success:
   1485                     return data

/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py
in _try_get_data(self, timeout)
   1308         #   (bool: whether successfully get data, any: data if
successful else None)
   1309         try:
-> 1310             data = self._data_queue.get(timeout=timeout)
   1311             return (True, data)
   1312         except Exception as e:
```

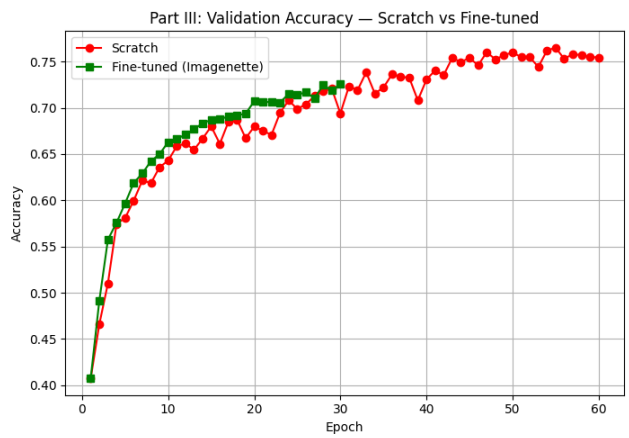
```
/usr/lib/python3.12/multiprocessing/queues.py in get(self, block,
timeout)
    120                 self._rlock.release()
    121         # unserialize the data after having released the lock
--> 122         return _ForkingPickler.loads(res)
    123
    124     def qsize(self):

/usr/local/lib/python3.12/dist-packages/torch/multiprocessing/reductio
ns.py in rebuild_storage_fd(cls, df, size)
    538
    539 def rebuild_storage_fd(cls, df, size):
--> 540     fd = df.detach()
    541     try:
    542         storage = storage_from_cache(cls, fd_id(fd))

/usr/lib/python3.12/multiprocessing/resource_sharer.py in detach(self)
     55         def detach(self):
     56             '''Get the fd.  This should only be called
once.'''
---> 57             with _resource_sharer.get_connection(self._id) as
conn:
     58                 return reduction.recv_handle(conn)
     59

/usr/lib/python3.12/multiprocessing/resource_sharer.py in
get_connection(ident)
     84         from .connection import Client
     85         address, key = ident
---> 86         c = Client(address,
authkey=process.current_process().authkey)
     87         c.send((key, os.getpid()))
     88         return c

/usr/lib/python3.12/multiprocessing/connection.py in Client(address,
family, authkey)
    517         c = PipeClient(address)
    518     else:
--> 519         c = SocketClient(address)
    520
    521     if authkey is not None and not isinstance(authkey, bytes):

/usr/lib/python3.12/multiprocessing/connection.py in
SocketClient(address)
    645     with socket.socket( getattr(socket, family) ) as s:
    646         s.setblocking(True)
--> 647         s.connect(address)
    648         return Connection(s.detach())
    649
```

```
FileNotFoundError: [Errno 2] No such file or directory
```

```
# ── Part III Comparison Plot ──
df_scratch   = pd.read_csv(logger_v3a.experiment.metrics_file_path)
df_finetune  = pd.read_csv(logger_v3b.experiment.metrics_file_path)

val_loss_scratch  = df_scratch.dropna(subset=['val_loss'])
['val_loss'].values
val_acc_scratch   = df_scratch.dropna(subset=['val_acc'])
['val_acc'].values
val_loss_ft       = df_finetune.dropna(subset=['val_loss'])
['val_loss'].values
val_acc_ft        = df_finetune.dropna(subset=['val_acc'])
['val_acc'].values

fig, axes = plt.subplots(1, 2, figsize=(14, 5))

axes[0].plot(range(1, len(val_loss_scratch)+1), val_loss_scratch, 'r-
o', label='Scratch')
axes[0].plot(range(1, len(val_loss_ft)+1),      val_loss_ft,      'g-
s', label='Fine-tuned (Imagenette)')
axes[0].set_title('Part III: Validation Loss — Scratch vs Fine-tuned')
axes[0].set_xlabel('Epoch')
axes[0].set_ylabel('Loss')
axes[0].legend()
axes[0].grid(True)

axes[1].plot(range(1, len(val_acc_scratch)+1), val_acc_scratch, 'r-o',
label='Scratch')
axes[1].plot(range(1, len(val_acc_ft)+1),      val_acc_ft,      'g-s',
label='Fine-tuned (Imagenette)')
axes[1].set_title('Part III: Validation Accuracy — Scratch vs Fine-
tuned')
axes[1].set_xlabel('Epoch')
axes[1].set_ylabel('Accuracy')
axes[1].legend()
axes[1].grid(True)

plt.tight_layout()
plt.savefig('part3_transfer_learning_comparison.png', dpi=150,
bbox_inches='tight')
plt.show()

print("\n" + "="*50)
print("FINAL RESULTS SUMMARY")
print("="*50)
print(f"Part I  — Imagenette (No Reg.)  Test Acc: {result_v1[0]
['test_acc']:.4f}")
print(f"Part II — Imagenette (With Reg.) Test Acc: {result_v2[0]
```

```
['test_acc']:.4f}")
print(f"Part III — CIFAR-10 Scratch      Test Acc: {result_scratch[0]
['test_acc']:.4f}")
print(f"Part III — CIFAR-10 Fine-tuned   Test Acc: {result_finetune[0]
['test_acc']:.4f}")
print("="*50)
```






```
==================================================
FINAL RESULTS SUMMARY
==================================================
Part I  — Imagenette (No Reg.)  Test Acc: 0.6531
Part II — Imagenette (With Reg.) Test Acc: 0.7819
Part III — CIFAR-10 Scratch      Test Acc: 0.7610
Part III — CIFAR-10 Fine-tuned   Test Acc: 0.7194
==================================================
```